INCH-POUND

MIL-I-25482D(USAF)
30 September 1993

SUPERSEDING
MIL-I-25482C(USAF)
18 DECEMBER 1986

MILITARY SPECIFICATION

INDICATOR, ENGINE TESTING

This specification is approved for use by the Department of the Air Force and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for one type of synchro style pressure ratio thrust indicator designated Type MR-2.

2.1 Government documents.

2.1.1 Specification, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

Federal

PPP-B-1672	Boxes, Shipping, Reusable With Cushioning

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Oklahoma City Air Logistics Center/TICLA, Tinker AFB OK 73145-3037 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 6620

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PPP-C-795	Cushioning Material, Packaging (Flexible Closed Cell Plastic Film For Long Shipping Cycle Applications).

Military

MIL-P-116	Preservation, Methods Of
MIL-B-121	Barrier Material, Greaseproofed, Waterproofed, Flexable
MIL-I-7057	Indicator, Synchro, Aircraft, General Specification For
MIL-B-27497	Bearing, Jewel, Sapphire Or Ruby, Synthetic

STANDARDS

Federal

FED-STD-101	Test Procedures For Packaging Materials
FED-STD-595	Colors

Military

MIL-STD-129	Marking For Shipment And Storage
MIL-STD-130	Identification Marking Of U.S. Military Property
MIL-STD-794	Parts And Equipment Procedures For Packaging Of
MIL-STD-2073-1	DOD Material Procedures For Development And Application Of Packaging Requirements
MS33585	Pointer, Dial, Standard Design Of Aircraft Instrument
MS33639	Case, Instrument, Clamp Mounted Aircraft
MS33678	Connector, Receptacle, Electric, Integral Mounting

(Unless otherwise indicated copies of federal and military specifications, standards, and handbooks are available from the Standardization Document Order Desk, Bldg 4D, 700 Robbins Ave, Philadelphia, PA 19111-5094).

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

AF Drawing 898894	Luminescent Material, Fluorescent

2.2 Order of precedence. In the event of a conflict between the test of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 Qualification. Indicators furnished under this specification shall be products which are authorized by the qualifying activity for listing on the applicable qualified products list at the time set for opening of bids (see 4.4 and 6.3).

3.2 General specification. The requirements of MIL-I-7057 apply as requirements of this specification with the exceptions and additions called out herein. When the two specifications conflict, this specification shall govern.

3.2.1 Recovered materials. Recovered materials shall be used to the maximum extent possible without jeopardizing the end use of the item.

3.3 Design and construction. The indicator shall be designed to indicate, at a remote station, the pressure ratio of a turbo-jet engine when used in conjunction with a pressure ratio transmitter wired in accordance with figure 2.

3.3.1 Case. The case shall conform to MS33639 and shall be hermetically sealed. The length of the case, excluding the connector, shall be 4.50 inches maximum.

3.3.1.1 Filling tube. The filling tube, if used, shall be at least 1.25 inches long and shall not be visible when installed in the aircraft.

3.3.2 Dual synchros. The indicator shall contain two self-synchronous synchros geared or otherwise connected to two separate pointers. There shall be no mechanical connection between the output shafts of the two synchros.

3.3.3 Dial. The dial shall be fabricated of metal and shall be securely attached to the frame of the indicator mechanism. The dial shall conform to figure 2.

3.3.3.1 Dial to cover glass distance. The distance between the indicator dial and the inside of the cover glass shall be the minimum practicable and shall not exceed 0.200 inch.

3.3.4 Pointer.

3.3.4.1 Main dial pointer. The pointer shall conform to MS33585-8, except that the taper at the point shall extend .3125 inch and the length of the pointer shall be such that the tip will overlap from 1/3 to 2/3 the length of the shortest graduation.

3.3.4.2 Sub dial pointer. The pointer shall conform to figure 2.

3.3.5 Markings.

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3.3.5.1 Dial and pointer finish. The following markings shall be finished in fluorescent-luminescent material conforming to AF Drawing 898894. The background of the dial and unshaded portion of the pointer shall be lusterless black, color number 37038 of FED-STD-595. The dimension of the markings shall be as listed below.

	Heights or Length Inch + .016	Width of Line or Graduation Inch + .005
Main Dial Numbers	.125	.020
Sub-Dial Numerals	.094	.016
Main Dial Graduation	.094	.016
Sub Dial Graduation	.063	.010
Lettering	.125	.020
Electrical Zero Reference	.031	.010
Shaded Portion of Pointer	-	-

3.3.6 Electrical receptacle. The electrical receptacle shall conform to MS33678R20-7P.

3.3.7 Wiring. Indicator internal and external wiring from indicator to transmitter shall be in accordance with figure 1.

3.3.8 Synthetic jewel bearings. When sapphire or ruby synthetic jewel bearings are used, they shall be in accordance with MIL-B-27497.

3.4 Weight. The weight of the indicator shall not exceed 1.0 pound.

3.5 Identification of product. Equipment, assemblies, and parts shall be marked for identification in accordance with MIL-STD-130.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the government. The government reserves the right to perform any of the inspection set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with

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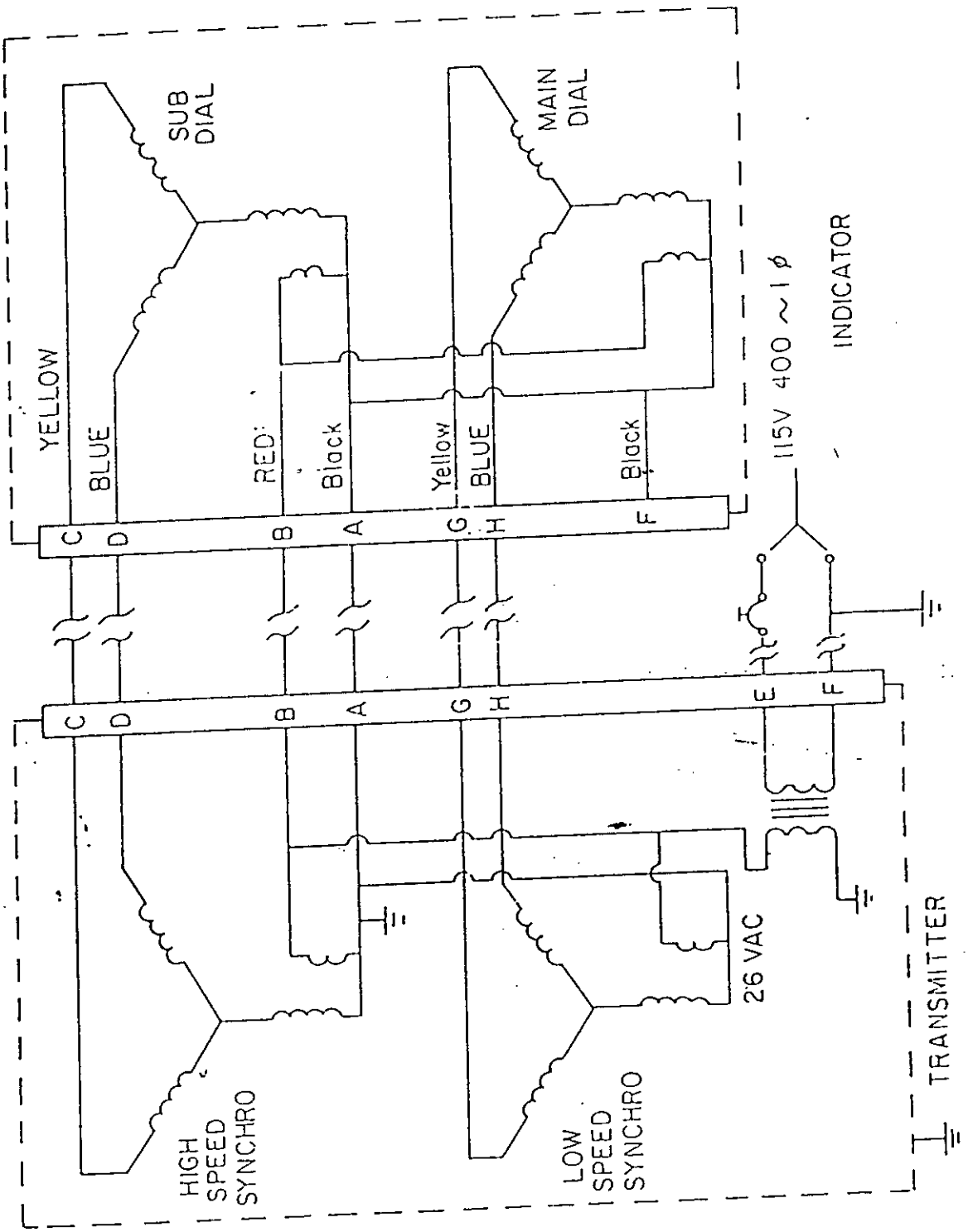


FIGURE 1 - Wiring diagram

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all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 General specification. The indicator shall be inspected and tested in accordance with MIL-I-7057 and as specified herein.

4.3 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.4)
b. Quality Conformance inspection (see 4.5)

4.4 Qualification inspection.

4.4.1 Test samples. The qualification test samples shall consist of three indicators representative of production equipment. The samples shall be identified with the manufacturer's part number and such other information as required by the qualifying activity.

4.4.2 Qualification tests. The qualification tests shall consist of all the tests of this specification including the test of MIL-I-7057.

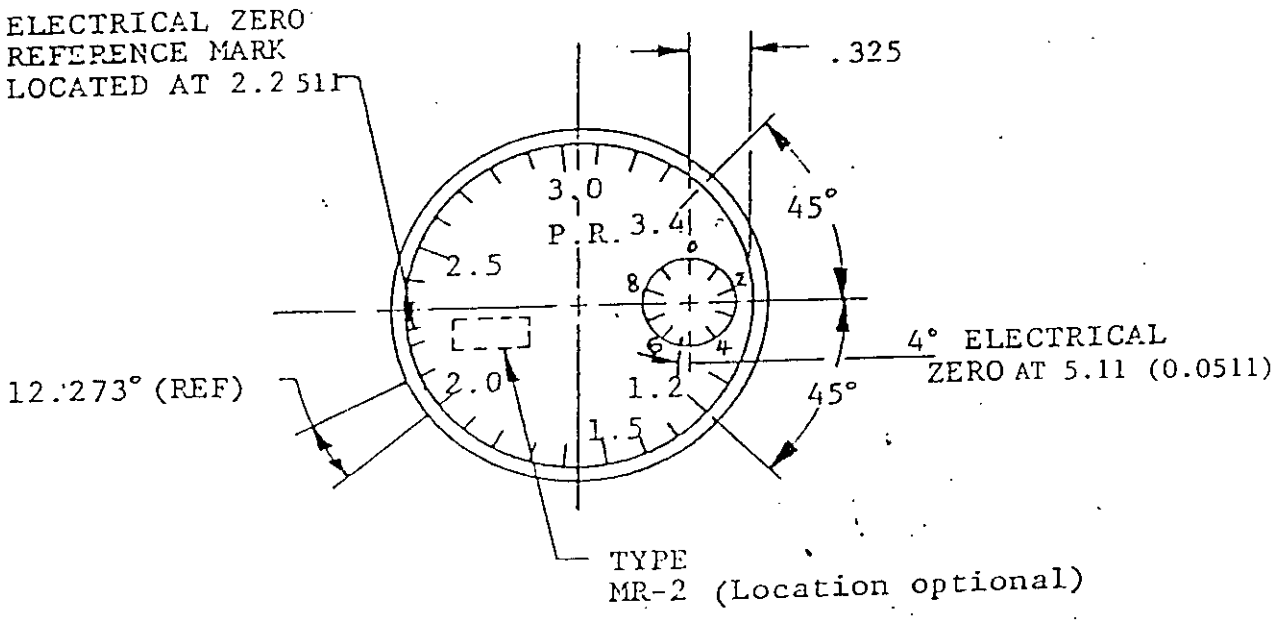
4.5 Quality conformance inspection. Quality conformance inspection shall consist of individual inspection and sampling inspection. Tests shall be in accordance with MIL-I-7057 and as specified herein. The test described under 4.6.7 shall replace the fogging test of MIL-I-7057.

4.6 Test methods.

4.6.1 Electrical zero. Connect the indicator to the electrical zero transformer per Figure 3. The indicator dials shall move to the electrical zero positions shown in Figure 2.

4.6.2 Scale error and friction. The indicator shall be tested for scale error and friction at the test points specified in table 1. Scale error and friction error shall not exceed the tolerances specified in Table II.

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DIMENSIONS IN INCHES.
UNLESS OTHERWISE SPECIFIED
TOLERANCES: DECIMALS ± 0.010
ANGLES ± 0° 10'

FIGURE 2. Details, dial and pointer.

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TABLE I. TEST POINTS.

Main Dial Low Speed		Sub-Dial High Speed	
Ratio	Degree	Ratio	Degrees
1.05	32.591		
1.10	38.727		
1.20	51.000	0.01	32.00
1.50	87.818	0.02	68.00
1.80	124.636	0.03	104.00
2.00	149.182	0.04	104.00
2.20	173.727	0.05	176.00
2.30	186.000	0.06	212.00
2.40	198.273	0.07	248.00
2.60	222.818	0.08	284.00
3.00	271.909	0.09	320.00
3.40	321.000	0.00	356.00

TABLE II. TOLERANCES.

Test	Pressure Main Dial	Ratio Units
Scale error	+ 0.020	+ 0.001
Room temp	+ 0.020	+ 0.001
High temp	+ 0.020	+ 0.001
Low temp	+ 0.006	+ 0.001
Position error		
Friction error		
Room temp	+ 0.010	+ 0.001
High temp	+ 0.010	+ 0.001
Low temp	+ 0.016	+ 0.001
Vibration		
Pointer oscillation	+ 0.006	+ 0.001
Pointer variation	+ 0.006	+ 0.001

4.6.3 Position error. The tolerance specified in Table II shall apply.

4.6.4 Vibration error. Pointer oscillation and variation shall not exceed the tolerances specified in Table II.

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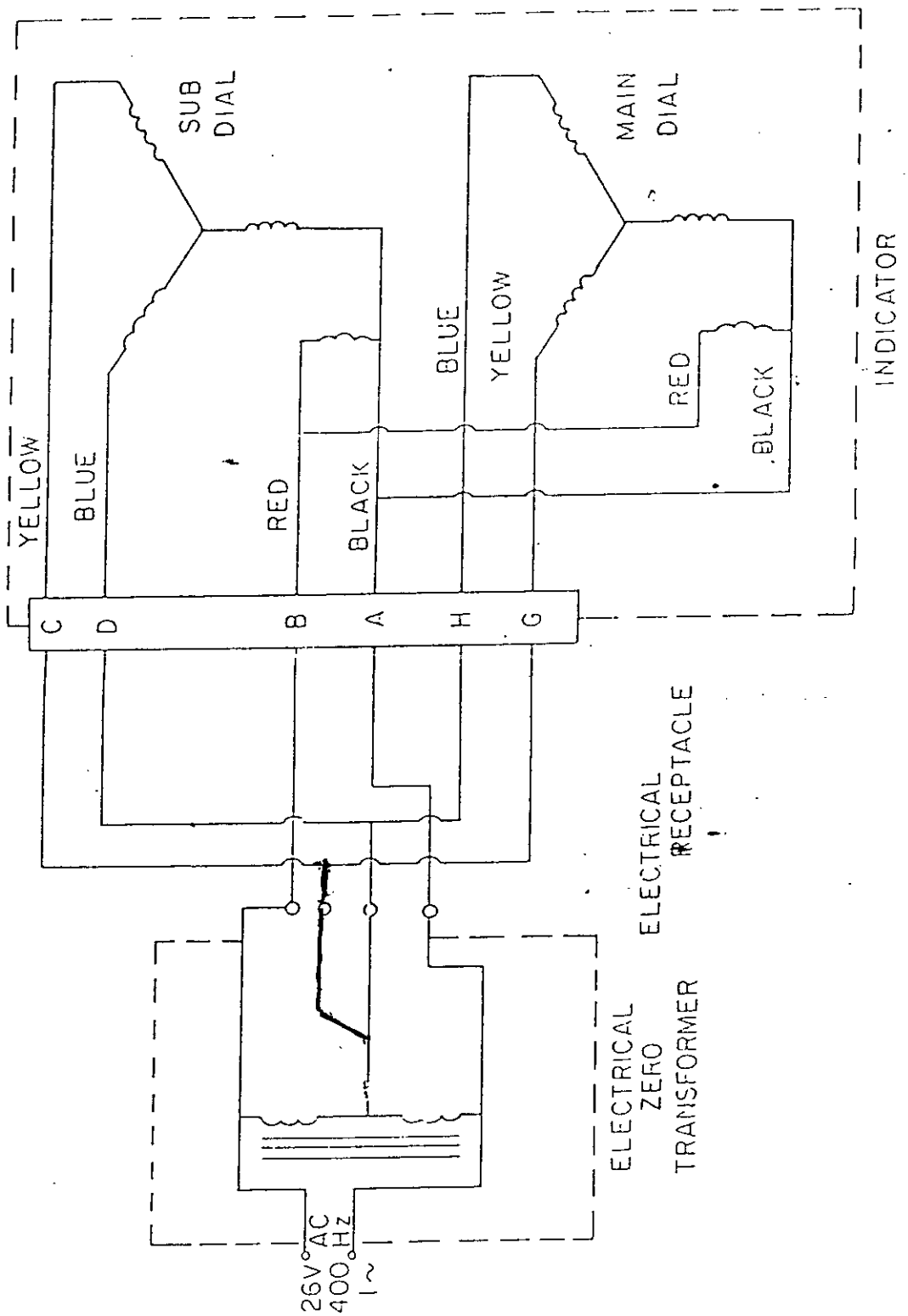


FIGURE 3 Electrical Zero Test Connections

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4.6.5 Low temperature. Scale and friction errors shall not exceed the tolerances specified in Table II. The indicator shall not require more than 5 seconds to come within the specified tolerance.

4.6.6 High temperature. Scale and friction errors shall not exceed the tolerances specified in Table II.

4.6.7 Fogging. The indicator shall be placed in a 90° + 5°C controlled ambient temperature for a minimum of 2 hours. The indicator shall then be removed from the high temperature chamber and within 30 seconds the indicator face shall be immersed to a depth of 1 to 2 inches in a water bath maintained at a temperature of 0° to 5°C. After 3 minutes the indicator shall be removed from the cold water and dried with air from a compressed air line. Any evidence of oil water fog inside the cover glass shall be cause for rejection.

4.6.8 High altitude-low temperature. The high altitude-low temperature test shall not be conducted.

4.6.9 Low temperature exposure. The low temperature and room temperature scale and friction error tolerances of Table II shall apply.

4.6.10 High temperature exposure. The high temperature and room temperature scale and friction error tolerances of Table II shall apply.

4.7 Inspection of preparation for delivery. Inspection of the preservation, packaging, and packing shall be conducted in accordance with the quality assurance provisions of MIL-STD-794 to insure, compliance with the requirements of section 5.

5. PACKAGING

5.1 Preservation. Preservation shall be level "A" or "C", as specified. (see 6.2)

5.1.1 Preservation-packaging. Unless otherwise specified by the contracting activity, item shall be packaged in Quantity Unit Pack (QUP) of one each. Each item will be provided a preservation method IC-1 In Accordance With (IAW) MIL-P-116.

5.1.2 Cleaning. Item shall be cleaned IAW MIL-P-116 C-1 process.

5.1.3 Drying. Immediately after cleaning, the item shall be dried following any one or combination of the drying procedures listed in MIL-P-116. The drying procedures employed shall not be injurious to the item.

5.1.4 Level A. Item shall be preserved IAW MIL-P-116 and MIL-STD-2073-1 to provide a method IC-1.

5.1.5 Level C. Item shall be individually preserved in a manner that will

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afford adequate protection against corrosion, deterioration, and physical damage during shipment from the supply source to the first receiving activity as specified in MIL-P-116 and MIL-STD-2073-1.

5.2 Packing. Packing shall be level "A", "B", or "C", as specified (see 6.2).

5.2.1 Container-cushioning. Unit container requirements shall conform to MIL-B-121, barrier material, unless otherwise specified by the contracting activity. Unit containers shall be large enough to allow for application of sufficient cushioning/wrap material, between the item and unit container. Item cushioning/wrap material shall conform to PPP-C-795 and provide sufficient density and thickness to protect the unit container from punctures. Intermediate/shipping containers shall be large enough to allow for application of sufficient cushioning material, between the unit container and intermediate/shipping container. Cushioning material shall be of sufficient density and thickness to insure shock transmission does not exceed peak values in G's established for the item.

5.2.2 Level A. Fiberboard containers do not meet Level "A" container criteria.

5.2.3 Level B. Item will be preserved as specified in 5.1 and shall be packed in exterior containers conforming to PPP-B-1672, weather-resistant unless otherwise specified by the contracting activity. Exterior container shall be uniform in shape, size and minimum tare and cube, consistent with the protection required. Closure shall be in accordance with appropriate PPP-B-1672 procedures, as specified by contractor, Special Packaging Instructions (SPI), and contracting activity.

5.2.4 Level C. Item will be packed in such a manner that will afford adequate protection against physical damage during direct domestic shipment from the supply source to the first receiving activity. These packs shall conform to MIL-STD-129.

5.3 Marking. Unit, intermediate, and exterior containers shall be marked IAW special markings required by the contracting activity, SPI and MIL-STD-129.

5.4 Inspection and test. Test of methods of preservation shall be accomplished in accordance with section 4 of MIL-P-116 to insure compliance with section 5 of this specification. Packaging tests shall be conducted IAW rough handling as specified in FED-STD-101.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The type MR-2 indicator covered by this specification is

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intended for use to indicate aircraft turbo-jet engine pressure ratio thrust measurements.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification
b. When sampling plan B tests will not be conducted (see MIL-I-7057)
c. Level of packaging and packing required
d. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1)

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are at the time set for opening of bids, qualified for inclusion in the application Qualified Products List whether or not such products have actually been so listed by the date. The attention of the suppliers is called to this requirement, and manufactures are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. The activity responsible for the Qualified Products List is Oklahoma City Air Logistics Center, TICLA, Tinker AFB, Oklahoma 73145-3037, and information pertaining to qualification of products may be obtained from that activity.

6.4 Subject term (key word) listing.

Aircraft jet engine
Dual synchros
Fluorescent-luminescent markings
Hermetically sealed
Length 4.50 inches maximum
Pressure ratio thrust

6.5 Changes from previous issue. Asterisks (or vertical lines) are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian:
Air Force - 99

Preparing activity:
Air Force - 71

Project No:
6620-F510

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-I-25482D(USAF)	93/09/30

3. DOCUMENT TITLE
INDICATOR, ENGINE TESTING

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code) (1) Commercial	(2) AUTOVON
OC-ALC/TICLA		336-5960
c. ADDRESS (Include Zip Code) RINKER AFB OK 73145-3037	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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